Factor XIa inhibition with asundexian after acute non-cardioembolic ischaemic stroke (PACIFIC-Stroke): an international, randomised, double-blind, placebo-controlled, phase 2b trial

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Summary

Background Asundexian (Bayer AG, Leverkusen, Germany), an oral small molecule factor XIa (FXIa) inhibitor, might prevent thrombosis without increasing bleeding. Asundexian's effect for secondary prevention of recurrent stroke is unknown.

Methods In this randomised, double-blind, placebo-controlled, phase 2b dose-finding trial (PACIFIC-Stroke), patients with acute (within 48 h) non-cardioembolic ischaemic stroke were recruited from 196 hospitals in 23 countries. Patients were eligible if they were aged 45 years or older, to be treated with antiplatelet therapy, and able to have a baseline MRI (either before or within 72 h of randomisation). Eligible participants were randomly assigned (1:1:1:1), using an interactive web-based response system and stratified according to anticipated antiplatelet therapy (single *vs* dual), to once daily oral asundexian (BAY 2433334) 10 mg, 20 mg, or 50 mg, or placebo in addition to usual antiplatelet therapy, and were followed up during treatment for 26–52 weeks. Brain MRIs were obtained at study entry and at 26 weeks or as soon as possible after treatment discontinuation. The primary efficacy outcome was the dose–response effect on the composite of incident MRI-detected covert brain infarcts and recurrent symptomatic ischaemic stroke at or before 26 weeks after randomisation. The primary safety outcome was major or clinically relevant non-major bleeding as defined by International Society on Thrombosis and Haemostasis criteria. The efficacy outcome was assessed in all participants assigned to treatment, and the safety outcome was assessed in all participants who received at least one dose of study treatment. This study is registered with ClinicalTrials.gov, NCT04304508, and is now complete.

Findings Between June 15, 2020, and July 22, 2021, 1880 patients were screened and 1808 participants were randomly assigned to asundexian 10 mg (n=455), 20 mg (n=450), or 50 mg (n=447), or placebo (n=456). Mean age was 67 years (SD 10) and 615 (34%) participants were women, 1193 (66%) were men, 1505 (83%) were White, and 268 (15%) were Asian. The mean time from index stroke to randomisation was 36 h (SD 10) and median baseline National Institutes of Health Stroke Scale score was 2·0 (IQR 1·0–4·0). 783 (43%) participants received dual antiplatelet treatment for a mean duration of 70·1 days (SD 113·4) after randomisation. At 26 weeks, the primary efficacy outcome was observed in 87 (19%) of 456 participants in the placebo group versus 86 (19%) of 455 in the asundexian 10 mg group (crude incidence ratio 0·99 [90% CI 0·79–1·24]), 99 (22%) of 450 in the asundexian 20 mg group (1·15 [0·93–1·43]), and 90 (20%) of 447 in the asundexian 50 mg group (1·06 [0·85–1·32]; t statistic –0·68; p=0·80). The primary safety outcome was observed in 11 (2%) of 452 participants in the placebo group versus 19 (4%) of 445 in the asundexian 10 mg group, 14 (3%) of 446 in the asundexian 20 mg group, and 19 (4%) of 443 in the asundexian 50 mg group (all asundexian doses pooled *vs* placebo hazard ratio 1·57 [90% CI 0·91–2·71]).

Interpretation In this phase 2b trial, FXIa inhibition with asundexian did not reduce the composite of covert brain infarction or ischaemic stroke and did not increase the composite of major or clinically relevant non-major bleeding compared with placebo in patients with acute, non-cardioembolic ischaemic stroke.

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Introduction

Non-cardioembolic ischaemic strokes account for 75% of all ischaemic strokes.[1] Despite advances in stroke preventive therapies for patients with non-cardioembolic ischaemic stroke, the recurrence rate is substantial, averaging more than 6% in the year after stroke despite

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Research in context

Evidence before this study

Antiplatelet therapy is the guideline-recommended antithrombotic prophylaxis for secondary prevention of non-cardioembolic ischaemic stroke. The benefit of anticoagulation therapy in this setting has not been established. Dual pathway inhibition combining an anticoagulant with an antiplatelet agent is hypothetically appealing, but this approach has not been explored to date due to the absence of safe oral anticoagulants. Factor XIa (FXIa) inhibitors, such as asundexian, might prevent thrombosis without increasing bleeding.

Added value of this study

This is the first randomised placebo-controlled trial to assess the effect of FXIa inhibition, using asundexian, when added to antiplatelet therapy for secondary prevention of non-cardioembolic ischaemic stroke. No difference was seen between asundexian (10 mg, 20 mg, and 50 mg daily) versus placebo on the primary efficacy outcome of composite symptomatic recurrent ischaemic stroke and incident covert brain infarcts. And no difference was seen between asundexian and placebo for the primary safety endpoint of the composite of major bleeding and clinically relevant non-major bleeding, according to the criteria of the International Society on Thrombosis and Haemostasis. However, by post-hoc analysis, inhibition of FXIa with asundexian reduced the composite of recurrent ischaemic stroke and transient ischaemic attack compared with placebo in patients with acute, non-cardioembolic ischaemic stroke, particularly for those with atherosclerosis, without increasing bleeding.

Implications of all the available evidence

Although we found no difference between asundexian and placebo for the primary efficacy or safety endpoints, the post-hoc results from this phase 2 trial are promising and require independent validation in an adequately powered phase 3 randomised trial before being applied to clinical care for secondary stroke prevention.

guideline-recommended treatment; therefore, more effective preventive treatments are an important unmet need.[2–4] In addition to clinical stroke recurrence, there is a substantial burden of covert brain infarction during the year after stroke. Generally, covert brain infarcts have been associated with cognitive and functional decline.[5] Guideline-recommended antithrombotic prophylaxis of patients who have non-cardioembolic ischaemic stroke includes long-term single antiplatelet therapy, sometimes after short-term dual antiplatelet therapy.[1,6,7] To date, no clinical studies have established a benefit of anticoagulation for secondary prevention in patients with non-cardioembolic ischaemic stroke. However, dual pathway antithrombotic therapy (ie, combining antiplatelet therapy with anticoagulant therapy) substantially reduced ischaemic stroke recurrence compared with aspirin alone in patients with stable coronary and peripheral artery atherosclerosis in the COMPASS trial.[8,9]

Asundexian is an oral direct inhibitor of activated coagulation factor XI (FXIa) that might have less associated bleeding than other available anticoagulants.[10] Patients with ischaemic stroke have increased levels of factor XI (FXI), and patients with inherited FXI deficiency have lower risks of stroke.[11–15] Several compounds targeting FXI or FXIa are in development, including small molecules, antibodies, and antisense approaches.[16] Phase 2 randomised trials have found effective prevention of thromboembolism and low risk of bleeding associated with use of FXI and FXIa inhibitors in patients undergoing total knee arthroplasty.[17–20] Therefore, asundexian is an attractive candidate for assessment as a potential add-on to antiplatelet therapy in patients with acute ischaemic stroke for secondary stroke prevention. The PACIFIC-Stroke trial is a phase 2b dose-finding study designed to assess the safety and potential efficacy of asundexian for prevention of brain infarction, both covert and symptomatic, in patients with acute non-cardioembolic ischaemic stroke. FXIa inhibition with asundexian added to usual antiplatelet therapy was hypothesised to be more efficacious than antiplatelet therapy alone for the prevention of recurrent symptomatic ischaemic stroke and incident covert brain infarcts in patients with acute non-cardioembolic ischaemic stroke and it was hypothesised that bleeding complications would not be increased by asundexian.

Methods

Study design and participants

In this international, randomised, placebo-controlled, double-blind, parallel-group, dose-finding phase 2b trial, participants were recruited from 196 hospitals in 23 countries in two stages (part A and part B). Individuals aged 45 years or older with non-cardioembolic ischaemic stroke (with persistent signs or symptoms that lasted for at least 24 h or acute brain infarction documented via brain imaging) were eligible for participation within 48 h of symptom onset (or last known to be without symptoms) if their treating physician intended to treat them with antiplatelet therapy during the study period, if they had brain imaging (CT or MRI) that eliminated haemorrhagic stroke or other pathology that could explain symptoms, were able to undergo a baseline MRI (either before or within 72 h after randomisation), and willing to adhere to study procedures. Furthermore, in part A of recruitment, participants were required to have had a minor ischaemic stroke (ie, National Institutes of Health Stroke Scale [NIHSS] score of ≤7 at randomisation), which was later broadened in part B to

additionally include moderate ischaemic strokes (NIHSS score of 8–15).

The key exclusion criteria were previous ischaemic stroke within 30 days of index event, history of atrial fibrillation or suspicion of cardioembolic stroke, active bleeding or history of major bleeding, uncontrolled hypertension, estimated glomerular filtration rate of less than 30 mL/min per 1·73 m², clinically significant liver disease, major surgery within 30 days before randomisation or planned surgery or intervention during the study period, treatment with a strong inducer or inhibitor of cytochrome P450 isoenzyme 3A4 within 14 days of randomisation, and indication for full dose and long-term anticoagulation therapy during study conduct (appendix pp 41–43). Furthermore, eligibility for part A was restricted to patients who did not receive thrombolysis or endovascular thrombectomy after their index stroke, whereas in part B, those treated with thrombolysis or endovascular therapy, or both, could be included if the therapy occurred at least 24 h before randomisation.

Transition from part A to part B occurred after recruitment of the first 829 participants, after recommendation from the independent data monitoring committee after a pre-planned review of safety data.

The trial was conducted in accordance with the protocol (appendix pp 14–99). The protocol was approved by the relevant health authorities and the institutional review boards at each trial site. Written informed consent was obtained from each participant or their legally authorised representative.

Randomisation and masking

Eligible patients were randomly assigned (1:1:1:1) to receive either asundexian 10 mg, 20 mg, or 50 mg or placebo, in addition to required background antiplatelet therapy chosen by the local investigator. Randomisation was done using an interactive web-based response system that was blocked from previewing and stratified according to whether the participants were anticipated to receive single or dual antiplatelet therapy during study follow-up. All doses of asundexian tablets and placebo were identical in appearance. Investigators and participants were masked to treatment assignment during the trial.

Procedures

The index stroke of each participant was classified by local investigators by applying the Trial of Org 10172 in Acute Stroke Treatment (TOAST) classification scheme.[21]

Participants were given either asundexian (BAY 2433334) orally once daily or matched placebo for a period of 26–52 weeks.

All participants were required to undergo at least two MRIs that met study requirements. In addition to the baseline MRI (obtained after the index stroke either before randomisation or within 72 h after randomisation), participants underwent a final study MRI at 26 weeks or as soon as possible after early termination of assigned study treatment. Participants recruited during part A additionally had a study MRI at 13 weeks, which was used for safety assessment for the transition to part B, but not for identification of the primary outcome unless no later study MRI was available. MRI data were transmitted to the central MRI core laboratory at the University of Calgary (Calgary, AB, Canada) for independent interpretation by two radiologists who were masked to participant data and treatment assignment, with differences resolved by consensus. Incident brain infarcts were identified by direct comparison of the baseline MRI with the follow-up study MRI; assessing radiologists were required to have good inter-rater agreement for brain infarcts (kappa value ≥0·60) on an independent test set of 50 brain MRIs.

Patients were followed up for at least 26 weeks and up to 52 weeks and assessed for the occurrence of safety, efficacy, and adverse events at 2, 4, 8, 13, 20, 26, 32, 39, 46, and 52 (end of treatment) weeks after randomisation, and again 2 weeks after the end of treatment visit. Adherence to assigned therapy was assessed by means of interview and pill count at each clinic visit.

Investigators reviewed laboratory test results and assessed participants for the occurrence of bleeding and adverse events at each study visit. Participants diagnosed with atrial fibrillation after randomisation had protocol-mandated discontinuation of study drug with subsequent antithrombotic management determined by local guidelines; these patients continued to be followed up and were included in the intention-to-treat analyses. In the event of a temporary interruption of study drug for any reason, study drug was restarted as soon as medically justified in the opinion of the investigator. There was no predefined maximum limit for temporary treatment interruption.

Outcomes

The primary efficacy endpoint was the composite of symptomatic recurrent ischaemic stroke and incident covert brain infarcts detected on follow-up MRI at or before 26 weeks after randomisation. Incident covert brain infarcts were counted in participants without recurrent ischaemic stroke before repeat MRI to avoid counting new lesions related to symptomatic infarcts. We elected to assess the effect of different doses of asundexian on prevention of all types of brain infarction. The spectrum of brain infarction included both clinical recurrent strokes (requiring recognised symptoms or signs) and incident covert brain infarction (in the absence of recognised symptoms or signs). We acknowledge that prevention of covert brain infarcts has not been proven for antithrombotic prophylaxis. Incident covert brain infarcts were anticipated to be more frequent than recurrent clinical ischaemic strokes and were included to allow adequate power to assess superior efficacy in this phase 2b dose-finding trial. Additionally, we prospectively planned to look at the individual components of the primary efficacy outcome (ie, symptomatic ischaemic stroke at 26 weeks after

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See **Online** for appendix

randomisation and covert brain infarcts detected on follow-up MRI at or before 26 weeks after randomisation) as pre-planned secondary endpoints.

Further secondary efficacy endpoints that are reported as of end of study were the composite of ischaemic stroke, cardiovascular death, and myocardial infarction; symptomatic recurrent ischaemic stroke; any recurrent stroke (symptomatic ischaemic and haemorrhagic stroke); disabling stroke (defined as a modified Rankin Scale [mRS] score of ≥4); and all-cause mortality. Additionally, the secondary endpoint of the composite of recurrent symptomatic ischaemic stroke, covert brain infarcts detected on MRI, cardiovascular death, myocardial infarction, and systemic embolism was reported at 26 weeks.

The primary safety outcome was the composite of major bleeding and clinically relevant non-major bleeding, according to the criteria of the International Society on Thrombosis and Haemostasis (ISTH).[22] Secondary safety outcomes were all bleeding, ISTH-defined major bleeding, ISTH-defined clinically relevant non-major bleeding, ISTH-defined minor bleeding, and symptomatic intracerebral haemorrhage (non-traumatic). Additional prespecified exploratory safety outcomes included haemorrhagic transformation within the qualifying stroke detected by baseline MRI performed post-study drug or on follow-up study MRIs based on the Heidelberg classification of severity (haemorrhagic infarct 1 or 2 or parenchymal haematoma 1 or 2).[23] Further exploratory safety endpoints, as stipulated in the protocol (appendix pp 57–58), will be reported elsewhere.

Exploratory post-hoc efficacy outcomes were assessment of the effect of each asundexian dose on the occurrence of transient ischaemic attack and the composite of recurrent symptomatic ischaemic stroke and transient ischaemic attack.

Outcome events were reported by local physician investigators through electronic submission of case report forms. A central clinical events committee, whose members were unaware of treatment assignment, applied the protocol definitions and adjudicated all strokes, transient ischaemic attacks, myocardial infarctions, deaths, and bleeding events.

Definitions of outcome events are in the protocol (appendix pp 53–58). Transient ischaemic attack was defined as abrupt onset of a focal neurological deficit attributed to brain or retinal ischaemia with resolution of symptoms and signs within 24 h and without neuroimaging evidence of acute infarction.

Statistical analysis

The primary efficacy analysis was the assessment of the overall dose–response effect of asundexian on the primary efficacy outcome at 26 weeks. We planned to enrol 1800 participants (450 per intervention group) to have at least 80% power (one-sided $\alpha=0\cdot10$) to detect a dose–response association, on the basis of a maximum relative risk reduction of 25% with the highest dose (50 mg) of asundexian compared with placebo and an estimated rate of the primary outcome of 22·5% at 26 weeks among patients assigned to placebo. All efficacy outcomes were analysed using the intention-to-treat principle, such that the efficacy analyses included all patients randomly assigned to treatment who were alive at the time of analysis and regardless of treatment discontinuation.

We used multiple comparison procedures with modelling techniques for the primary efficacy analysis, allowing for the flexibility of modelling for dose estimation, while preserving the robustness to model misspecification associated with multiple comparison procedures. Based on these models and the observed data, we calculated optimal contrasts and the corresponding critical value. This analysis was done using the estimated covariance matrix of a logistic regression model. Based on the optimal contrast and the critical values, we used a one-sided test with $\alpha=0\cdot10$ based on the maximum value of the test statistics for the models in the candidate set (appendix pp 66–67). This method takes multiplicity into account. Thus, no further multiplicity adjustments were done. We calculated crude incidence ratios at 26 weeks for outcomes that required MRIs—ie, the composite primary efficacy outcome, and the secondary outcomes of its components—and their corresponding two-sided 90% confidence intervals. We estimated cause-specific hazard ratios (HRs) and their corresponding two-sided 90% CIs, at the end of the study using separate Cox proportional hazard models comparing each asundexian dose versus placebo for secondary efficacy outcomes that did not include covert brain infarcts on MRI. We calculated both the crude incidence ratio at 26 weeks and cause-specific HRs at the end of the study for the outcome of recurrent symptomatic ischaemic stroke. The cause-specific Cox proportional hazard model was based on the time to first occurrence of an event, with treatment as the independent variable. Imputation rules and pre-planned sensitivity analyses to account for missing values in patients without serial MRI data and recurrent symptomatic ischaemic stroke to fulfil the primary efficacy outcome are provided in the statistical analysis plan (appendix pp 125–26). We used Kaplan-Meier estimates to plot the cumulative-incidence risk of recurrent symptomatic ischaemic stroke over time.

We did exploratory analyses of treatment effects on the primary binary efficacy outcome at 26 weeks using unadjusted logistic regression models in prespecified subgroups (appendix pp 130–31). We did post-hoc exploratory subgroup analyses confined to the effect of asundexian 50 mg daily versus placebo on the composite outcome of recurrent symptomatic ischaemic stroke and transient ischemic attack (data driven) in prespecified subgroups indicating atherosclerotic disease (a priori on the basis of the COMPASS trial results).[8]

Safety endpoints were treatment emergent and assessed in all participants who received at least one dose of study treatment. For the primary safety analysis, we used a Cox

proportional hazard model to compare the proportion of pooled primary safety outcome events from all asundexian treatment groups with the proportion in the placebo group. The cause-specific Cox proportional hazard model was based on the time to first occurrence of an event, with treatment as the independent variable. All events from first intake of study drug up to 2 days after permanent study drug discontinuation were counted. We then replicated these analyses for the comparison of each dose of asundexian with placebo. We used Kaplan-Meier estimates to plot the cumulative-incidence risk of the primary safety outcome over time. We calculated the proportion, crude incidence ratios, and accompanying 90% CIs for the prespecified exploratory safety outcomes of haemorrhagic infarction and parenchymal haematoma in patients who had their baseline MRI done after initiation of study drug (up to 72 h after randomisation). Separately, we did the same calculations for new regions of haemorrhagic transformation seen on follow-up MRIs at 26 weeks in patients without haemorrhagic transformation of the index stroke on their baseline MRI.

Our methods for handling missing data are described in the protocol and statistical analysis plan (appendix pp 76, 112)

We considered p values of less than 0·10 to be significant for analyses of study outcomes. Additionally, we used descriptive statistics to compare baseline demographic and clinical variables between patients who underwent adequate serial study MRIs versus those who did not. p values of less than 0.05 were considered statistically significant in these descriptive analyses.

We did all statistical analyses using SAS (release 9.4). An independent data and safety monitoring committee monitored safety data on an ongoing basis. No formal interim analyses for efficacy were planned or done. This study is registered with ClinicalTrials.gov, NCT04304508.

Role of the funding source

The trial was initiated and funded by the trial sponsor, Bayer, who designed the trial with input from collaborators at the Population Health Research Institute (PHRI) at McMaster University (Hamilton, ON, Canada) and the trial steering committee. The PHRI identified steering committee members, proposed trial sites, collected and managed data, and performed the data analysis with input from the sponsor. The trial sponsor selected study investigators, provided trial medications, contracted with and paid the trial investigators, and undertook local site monitoring. The sponsor did not have a role in data collection or statistical analysis of the data. The sponsor reviewed and provided comments on the manuscript, but sponsor approval was not required for submission.

Results

Between June 15, 2020, and July 22, 2021, of 1880 people screened, 1808 participants were randomly assigned to

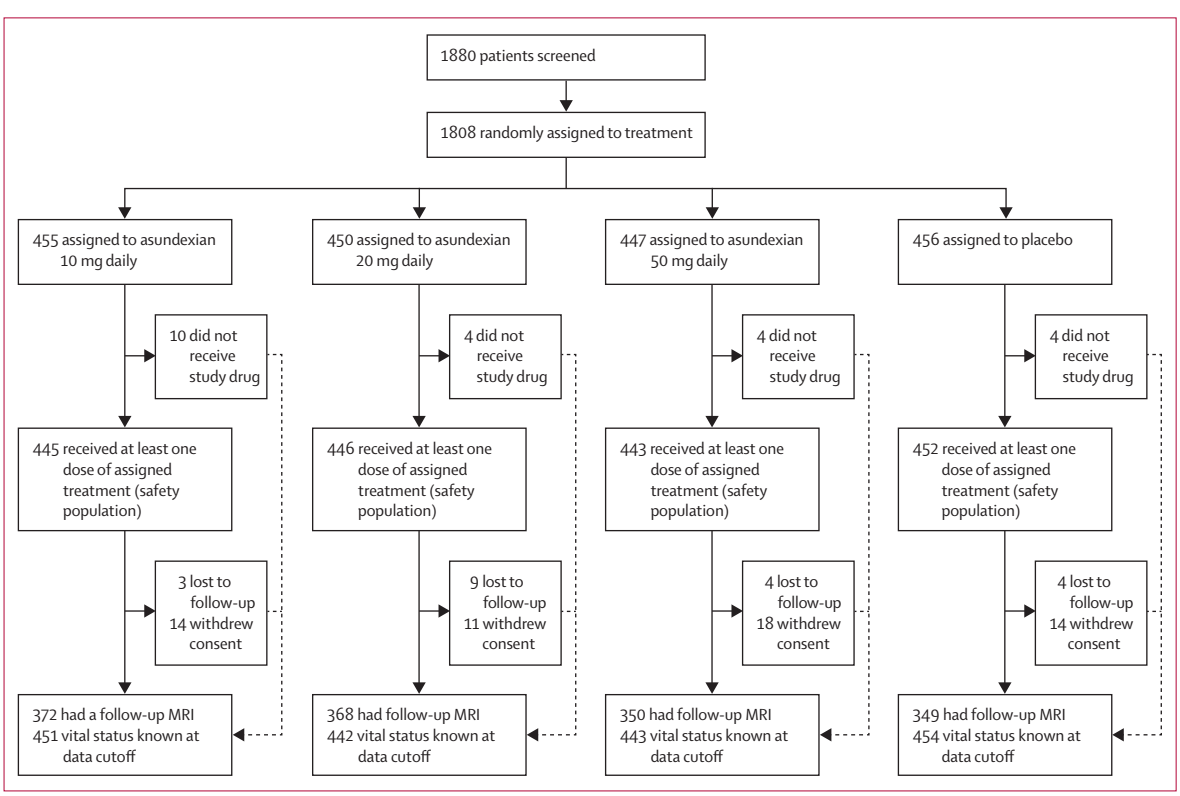

Figure: **Trial profile**

	Total (n=1808)	Placebo group (n=456)	Asundexian 10 mg group (n=455)	Asundexian 20 mg group (n=450)	Asundexian 50 mg group (n=447)
Age, years	67·0 (9·9)	66·6 (10·1)	66·8 (10·1)	67·6 (9·4)	67·0 (10·0)
Sex					
Female	615 (34%)	150 (33%)	161 (35%)	150 (33%)	154 (34%)
Male	1193 (66%)	306 (67%)	294 (65%)	300 (67%)	293 (66%)
Self-reported race					
White	1505 (83%)	380 (83%)	381 (84%)	377 (84%)	367 (82%)
Black	18 (1%)	3 (1%)	4 (1%)	3 (1%)	8 (2%)
Asian	268 (15%)	66 (14%)	67 (15%)	67 (15%)	68 (15%)
Other	3 (<1%)	1 (<1%)	0	1 (<1%)	1 (<1%)
Geographical region					
North America	23 (1%)	6 (1%)	6 (1%)	6 (1%)	5 (1%)
Western Europe and Australia	1007 (56%)	255 (56%)	253 (56%)	251 (56%)	248 (55%)
Eastern Europe	524 (29%)	131 (29%)	132 (29%)	130 (29%)	131 (29%)
Asia	254 (14%)	64 (14%)	64 (14%)	63 (14%)	63 (14%)
Bodyweight, kg	78·9 (16·6)	79·2 (16·0)	78·5 (17·4)	78·6 (16·4)	79·2(16·5)
BMI, kg/m^2	27·5 (4·8)	27·4 (4·7)	27·4 (5·0)	27·4 (4·8)	27·6 (4·9)
Current tobacco use	484 (27%)	126 (28%)	123 (27%)	113 (25%)	122 (27%)
Previous medical conditions					
Hypertension	1392 (77%)	346 (76%)	366 (80%)	333 (74%)	347 (78%)
Diabetes	501 (28%)	127 (28%)	122 (27%)	137 (30%)	115 (26%)
Hyperlipidaemia	955 (53%)	246 (54%)	225 (49%)	228 (51%)	256 (57%)
Heart failure	104 (6%)	24 (5%)	28 (6%)	26 (6%)	26 (6%)
Coronary artery disease	167 (9%)	37 (8%)	43 (9%)	41 (9%)	46 (10%)
Myocardial infarction	81 (4%)	17 (4%)	23 (5%)	18 (4%)	23 (5%)
Percutaneous coronary intervention or angioplasty, or coronary artery bypass grafting	80 (4%)	19 (4%)	16 (4%)	23 (5%)	22 (5%)
Carotid endarterectomy or stenting	13 (1%)	2 (<1%)	3 (1%)	4 (1%)	4 (1%)
Peripheral artery disease	45 (2%)	13 (3%)	11 (2%)	9 (2%)	12 (3%)
Chronic kidney disease	93 (5%)	29 (6%)	19 (4%)	29 (6%)	16 (4%)
Previous stroke or transient ischaemic attack	285 (16%)	75 (16%)	77 (17%)	68 (15%)	65 (15%)
History of gastrointestinal bleeding	18 (1%)	7 (2%)	3 (1%)	2 (<1%)	6 (1%)
Qualifying stroke subtype*					
Large-artery atherosclerosis	320 (18%)	76 (17%)	82 (18%)	73 (16%)	89 (20%)
Small-vessel occlusion	817 (45%)	210 (46%)	205 (45%)	196 (44%)	206 (46%)
Stroke of other determined aetiology	45 (2%)	6 (1%)	15 (3%)	10 (2%)	14 (3%)
Stroke of undetermined aetiology	616 (34%)	162 (36%)	149 (33%)	168 (37%)	137 (31%)
Cardioembolism	9 (<1%)	2 (<1%)	3 (1%)	3 (1%)	1 (<1%)
Extracranial or intracranial atherosclerosis proximal to the qualifying stroke	607 (34%)	142 (31%)	150 (33%)	161 (36%)	154 (34%)
Carotid artery atherosclerosis identified by vascular imaging	650 (36%)	166 (36%)	162 (36%)	170 (38%)	152 (34%)
NIHSS score at randomisation†					
Mean (SD)	2·8 (2·2)	2·9 (2·2)	2·8 (2·3)	2·7 (2·1)	3·0 (2·3)
8–15	58 (3%)	17 (4%)	14 (3%)	11 (2%)	16 (4%)
Day 7 mRS score‡	1·0 (1·0–2·0)	1·0 (1·0–2·0)	1·0 (1·0–2·0)	1·0 (1·0–2·0)	1·0 (1·0–2·0)
Dual antiplatelet therapy before index event	39 (2%)	8 (2%)	13 (3%)	10 (2%)	8 (2%)
Dual antiplatelet therapy after index event	783 (43%)	199 (44%)	197 (43%)	195 (43%)	192 (43%)
Blood pressure at randomisation, mm Hg					
Systolic blood pressure	138 (14)	137 (14)	139 (14)	138 (14)	139 (14)
Diastolic blood pressure	79 (10)	79 (10)	79 (11)	79 (10)	79 (10)
eGFR, mL/min per 1·73 m^2§	79 (21)	79 (22)	79 (21)	79 (20)	78 (20)

(Table 1 continues on next page)

	Total (n=1808)	Placebo group (n=456)	Asundexian 10 mg group (n=455)	Asundexian 20 mg group (n=450)	Asundexian 50 mg group (n=447)
(Continued from previous page)					
Time from qualifying stroke to randomisation, h					
Median (IQR)	38 (30–44)	37 (29–44)	40 (30–45)	38 (30–44)	38 (29–45)
≤24 h	193 (11%)	49 (11%)	41 (9%)	52 (12%)	51 (11%)
Thrombolysis before randomisation	217 (12%)	59 (13%)	59 (13%)	48 (11%)	51 (11%)
Endovascular thrombectomy before randomisation	52 (3%)	11 (2%)	15 (3%)	12 (3%)	14 (3%)

Data are mean (SD), n (%), or median (IQR). eGFR=estimated glomerular filtration rate. mRS=modified Rankin Scale. NIHSS=National Institutes of Health Stroke Scale. *Trial of Org 10172 in Acute Stroke Treatment criteria[21] applied by the local investigators. †Scores on the NIHSS range from 0 to 42, with higher scores representing worse neurologic deficits. ‡Scores on the mRS range from 0 to 6, with higher scores representing worse functional deficits. §Reported by local laboratories.

Table 1: **Baseline participant characteristics**

either asundexian 10 mg (n=455), 20 mg (n=450), or 50 mg (n=447), or placebo (n=456; figure). The trial ended as planned on Feb 18, 2022 (last patient visit) and data cutoff was on March 10, 2022, 26 weeks after the recruitment target was achieved. Part B recruitment of participants started on March 2, 2021. Participants were followed up for a median of 46·1 weeks (IQR 35·2–53·8). 22 participants who were randomly assigned did not receive any study treatment (figure). Demographic and clinical characteristics were evenly distributed between treatment groups (table 1). The mean age of the participants was 67 years (SD 10), 615 (34%) were female, 1193 (66%) were male, 1505 (83%) were White, 268 (15%) were Asian, 18 (1%) were Black, and 1007 (56%) participants were recruited from western Europe and Australia. The mean time from index stroke to randomisation was 36 h (SD 10). 1483 (82%) participants had signs and symptoms of the index stroke that lasted at least 24 h, with the remainder showing acute brain infarction on neuroimaging. The most common index stroke subtype was small-vessel occlusion, followed by stroke of undetermined cause and extracranial or intracranial atherosclerosis proximal to the qualifying stroke. The mean NIHSS score at randomisation was 2·8 (SD 2·2) and the median score was 2·0 (IQR 1·0–4·0); the median day 7 mRS score was 1·0 (IQR 1·0–2·0). 783 (43%) participants received dual antiplatelet therapy for a mean duration of 70·1 days (SD 113·4) after randomisation. 217 (12%) participants were treated with thrombolysis and 52 (3%) with endovascular thrombectomy before randomisation (all enrolled during part B).

An adequate baseline MRI was available for 1780 (98%) participants and 1439 (80%) participants had adequate follow-up MRIs, such that adequate baseline and follow-up MRIs were available in 1428 (79%) participants, and the mean time between MRIs was 26 weeks (SD 3; reasons for not having an adequate MRI are listed in the appendix [p 1]). Participants without adequate serial MRIs (380 [21%] of 1808) were slightly older (p<0·001) and more often women (p=0·04; appendix p 1). Discontinuation of trial medication due to atrial fibrillation diagnosed after randomisation occurred in 79 patients (4%; 59 [4%] of 1352 assigned to asundexian and 20 [4%] of 456 assigned to placebo) after a median of 1·8 months (IQR 0·2–3·7). 19 (1%) participants were lost to follow-up after a mean of 3·5 months (SD 3·1), and an additional 57 (3%) withdrew consent for follow-up after a mean of 1·8 months (SD 1). Vital status was available at the end of trial for 1790 (99%) of the patients who had been randomly assigned to treatment and not withdrawn consent or been lost to follow-up.

At 26 weeks after randomisation, the primary composite efficacy outcome was observed in 87 (19%) of 456 participants in the placebo group versus 86 (19%) of 455 in the asundexian 10 mg group, 99 (22%) of 450 in the asundexian 20 mg group, and 90 (20%) of 447 in the asundexian 50 mg group. No significant difference was seen between the placebo group and any of the asundexian dose groups (table 2) and we observed no significant dose response (Emax2 model t statistic –0·68; p=0·80). We found no significant heterogeneity of treatment effects in the prespecified exploratory subgroup analyses after adjustment for multiple comparisons (appendix pp 3–11). 275 (76%) of 362 events that comprised the composite primary outcome were incident covert brain infarcts. Incident covert brain infarcts were identified in 275 participants (MRI detected in 219 [80%] participants, imputed in 56 [20%] participants); 153 (70%) of 219 MRI-detected covert brain infarcts were small (≤15 mm in diameter) and subcortical, while 68 (31%) were cortical or large (>15 mm). The frequency of incident covert brain infarcts was similar across treatment groups (table 2).

As of data cutoff, 102 recurrent symptomatic ischaemic strokes occurred: 28 (6%) in the placebo group, 26 (6%) in the asundexian 10 mg group, 26 (6%) in the 20 mg group, and 22 (5%) in the 50 mg group (table 2; appendix p 12). Prespecified secondary efficacy outcomes are presented in table 2, with no significant differences seen between the placebo group and any of the asundexian dose groups. In post-hoc analysis, the frequency of the composite outcome of recurrent ischaemic stroke and

	Placebo (n=456)	Asundexian 10 mg group (n=455)	Asundexian 10 mg *vs* placebo	Asundexian 20 mg group (n=450)	Asundexian 20 mg *vs* placebo	Asundexian 50 mg group (n=447)	Asundexian 50 mg *vs* placebo
Primary outcome							
Ischaemic stroke or covert infarcts*	87 (19%)	86 (19%)	0·99 (0·79–1·24)	99 (22%)	1·15 (0·93–1·43)	90 (20%)	1·06 (0·85–1·32)
Secondary outcomes							
Components of the primary outcome*							
Incident covert brain infarcts on MRI†	64 (14%)	63 (14%)	0·99 (0·75–1·30)	74 (16%)	1·17 (0·90–1·51)	74 (17%)	1·17 (0·91–1·52)
Recurrent symptomatic ischaemic stroke*	23 (5%)	24 (5%)	1·05 (0·66–1·67)	25 (6%)	1·10 (0·69–1·75)	17 (4%)	0·75 (0·45–1·26)
Efficacy outcomes‡							
Recurrent symptomatic ischaemic stroke§	28 (6%)	26 (6%)	0·93 (0·59–1·45)	26 (6%)	0·94 (0·60–1·47)	22 (5%)	0·80 (0·50–1·27)
Any recurrent stroke§	30 (7%)	26 (6%)	0·86 (0·56–1·34)	26 (6%)	0·88 (0·56–1·36)	25 (6%)	0·85 (0·54–1·32)
Disabling stroke (mRS score of ≥4)§	3 (1%)	5 (1%)	1·67 (0·50–5·55)	5 (1%)	1·69 (0·51–5·62)	1 (<1%)	0·34 (0·05–2·27)
Recurrent symptomatic ischaemic stroke, vascular death, or myocardial infarction§	35 (8%)	33 (7%)	0·94 (0·63–1·40)	30 (7%)	0·87 (0·58–1·30)	33 (7%)	0·96 (0·64–1·43)
Recurrent symptomatic ischaemic stroke, incident covert brain infarct on MRI, cardiovascular death, myocardial infarction and systemic embolism*	79 (17%)	80 (18%)	0·95 (0·76–1·20)	87 (19%)	1·06 (0·85–1·33)	81 (18%)	1·03 (0·82–1·30)
All-cause mortality§	10 (2%)	10 (2%)	1·00 (0·48–2·09)	6 (1%)	0·60 (0·26–1·41)	17 (4%)	1·72 (0·89–3·32)
Post-hoc exploratory outcomes‡							
Transient ischaemic attack	11 (2%)	10 (2%)	0·91 (0·44–1·87)	2 (<1%)	0·18 (0·05–0·64)	2 (<1%)	0·18 (0·05–0·65)
Recurrent symptomatic ischaemic stroke or transient ischaemic attack	38 (8%)	35 (8%)	0·92 (0·63–1·35)	28 (6%)	0·74 (0·49–1·12)	24 (5%)	0·64 (0·41–0·98)

Data are n (%) or hazard ratio with 90% CI in parentheses, or crude incidence ratio with 90% CI in parentheses. The cause-specific Cox proportional hazard model is modelled on the basis of the time to first occurrence of the event. The independent variable is treatment. Hazard ratios are calculated separately for all comparisons. mRS=modified Rankin Scale. *Proportion of outcomes at 26 weeks and accompanying crude incidence ratios for these binary event outcomes are presented. †Incident covert brain infarct data missing in 352 patients and imputed in patients who did not otherwise meet the primary efficacy outcome based on having a symptomatic recurrent ischemic stroke. ‡Hazard ratios calculated using Cox proportional hazard model are presented. §Proportion of outcomes at end study.

Table 2: **Efficacy outcomes**

transient ischaemic attack was lower among participants assigned to asundexian than among participants assigned to placebo, particularly among those assigned to asundexian 50 mg (table 2). Of the 25 patients diagnosed with transient ischaemic attack, 14 (56%) had an MRI and seven (28%) had only CT for evaluation of the event.

A large, non-significant, relative risk reduction in recurrent symptomatic ischaemic stroke was observed with asundexian 50 mg daily among the prespecified subgroup of patients with stroke attributed to large-artery atherosclerotic disease (seven [8%] of 89 in the asundexian 50 mg group *vs* 11 [15%] of 76 in the placebo group; HR 0·53 [90% CI 0·24–1·17]). Similar reductions were seen when considering patients with imaging evidence of any extracranial or intracranial atherosclerosis supplying the qualifying infarct (six [3%] of 195 in the asundexian 50 mg group *vs* 12 [6%] of 198 in the placebo group; HR 0·52 [90% CI 0·23–1·18]), for the composite post-hoc outcome of recurrent ischaemic stroke and transient ischaemic attack in patients with qualifying stroke due to large-artery atherosclerotic disease (eight [9%] of 89 in the asundexian 50 mg group *vs* 12 [16%] of 76 in the placebo group; HR 0·56 [90% CI 0·26–1·19]), and those with any degree of extracranial or intracranial atherosclerosis (six [3%] of 195 in the asundexian 50 mg group *vs* 16 [8%] of 198 in the placebo group; HR 0·39 [90% CI 0·18–0·85]).

The primary safety composite outcome of ISTH major and clinically relevant non-major bleeding occurred while on treatment in 11 (2%) of 452 participants taking placebo, 19 (4%) of 445 taking asundexian 10 mg, 14 (3%) of 446 taking asundexian 20 mg, and 19 (4%) of 443 taking asundexian 50 mg (table 3; appendix p 13). We found no dose–response association and no significant increase in the proportion of primary safety outcome events in the pooled asundexian groups compared with the proportion of events in the placebo group (HR 1·57 [90% CI 0·91–2·71]; table 3). Primary intracerebral haemorrhages occurred in three (1%) participants who received asundexian 50 mg daily and one (<1%) who received placebo (HR 3·05 [90% CI 0·46–20·4]; table 3). On baseline MRIs done after study drug initiation, there was no increase in secondary haemorrhagic transformation of the index stroke resulting in haemorrhagic infarcts, including incident parenchymal haematoma 1 or parenchymal haematoma 2 among those assigned to asundexian (table 3).

Discussion

In this international, phase 2b dose-finding trial of patients with acute non-cardioembolic ischaemic stroke receiving antiplatelet therapy, we found no overall dose–response effect with asundexian on the primary efficacy composite outcome of recurrent ischaemic stroke and incident covert brain infarcts. This finding was in

	Placebo group (n=452)	Asundexian 10 mg group (n=445)	Asundexian 10 mg *vs* placebo	Asundexian 20 mg group (n=446)	Asundexian 20 mg *vs* placebo	Asundexian 50 mg group (n=443)	Asundexian 50 mg *vs* placebo	Asundexian all doses (n=1334)	Asundexian all doses *vs* placebo
Primary safety outcome*									
ISTH-defined major and clinically relevant non-major bleeding	11 (2%)	19 (4%)	1·71 (0·91–3·18)	14 (3%)	1·27 (0·66–2·47)	19 (4%)	1·74 (0·93–3·24)	52 (4%)	1·57 (0·91–2·71)
Secondary safety outcomes*									
All bleeding	44 (10%)	37 (8%)	0·82 (0·57–1·18)	48 (11%)	1·11 (0·79–1·56)	48 (11%)	1·10 (0·78–1·54)	133 (10%)	1·01 (0·76–1·34)
ISTH-defined major bleeding	4 (1%)	4 (1%)	0·98 (0·31–3·15)	3 (1%)	0·76 (0·22–2·66)	7 (2%)	1·76 (0·63–4·94)	14 (1%)	1·17 (0·46–2·96)
ISTH-defined clinically relevant non-major bleeding	7 (2%)	15 (3%)	2·11 (0·99–4·48)	12 (3%)	1·71 (0·78–3·75)	12 (3%)	1·72 (0·79–3·76)	39 (3%)	1·85 (0·94–3·64)
ISTH-defined minor bleeding	34 (8%)	21 (5%)	0·60 (0·38–0·95)	39 (9%)	1·17 (0·79–1·72)	34 (8%)	1·01 (0·67–1·50)	94 (7%)	0·92 (0·66–1·28)
Intracerebral haemorrhage	1 (<1%)	0	··	0	··	3 (1%)	3·05 (0·46–20·4)	3 (<1%)	1·00 (0·15–6·70)
Exploratory safety outcomes†									
Haemorrhagic infarction 1 and 2 on baseline MRI done after first dose of study drug (up to 72 h after randomisation)‡	93/296 (31%)	82/277 (30%)	0·94 (0·77–1·16)	78/265 (29%)	0·94 (0·76–1·16)	84/277 (30%)	0·97 (0·79–1·19)	244/819 (30%)	0·95 (0·80–1·12)
Parenchymal haematoma 1 and 2 on baseline MRI done after first dose of study drug (up to 72 h after randomisation)‡	4/296 (1%)	3/277 (1%)	0·80 (0·23–2·79)	1/265 (<1%)	0·28 (0·04–1·75)	0/277	··	4/819 (<1%)	0·36 (0·11–1·15)
New haemorrhagic infarction 1 and 2 on follow-up MRI§	47/323 (15%)	50/319 (16%)	1·08 (0·79–1·47)	53/332 (16%)	1·10 (0·81–1·49)	56/320 (18%)	1·20 (0·89–1·62)	159/971 (16%)	1·13 (0·87–1·45)
New parenchymal haematoma 1 and 2 on follow-up MRI§	1/323 (<1%)	0/319	··	1/332 (<1%)	0·97 (0·10–9·93)	0/320	··	1/971 (<1%)	0·33 (0·03–3·40)

Data are n (%), n/N (%), or hazard ratio with 90% CI in parentheses, or crude incidence ratios for secondary safety outcomes requiring MRI. ISTH=International Society on Thrombosis and Haemostasis. *Hazard ratios calculated using Cox model are presented. †Proportion of outcomes up to 26 weeks and accompanying crude incidence ratios for these binary event outcomes are presented. ‡Among 1115 patients who had a baseline MRI after their first dose of study drug. §Among 1294 patients without haemorrhagic transformation of the index stroke on the baseline MRI.

Table 3: **Treatment-emergent safety outcomes**

part accounted for by the absence of reduction of incident covert brain infarcts that contributed 75% of primary outcome events. However, in post-hoc analyses, treatment with asundexian 50 mg daily reduced the clinical outcome of recurrent symptomatic ischaemic stroke and transient ischaemic attack, and this reduction was pronounced among participants with coexistent atherosclerosis, although these findings should be interpreted with caution due to their post-hoc nature. We found no significant differences between the placebo and asundexian treatment groups in the primary or secondary safety outcomes. The frequency of haemorrhagic infarcts (reflecting secondary bleeding into the qualifying brain infarction) detected by study MRIs done after initiation of study drug were similar in all treatment groups. Although the primary and secondary efficacy endpoints were negative, post-hoc analyses suggest that asundexian might be effective for prevention of the composite of recurrent ischaemic stroke and transient ischaemic attack when added to antiplatelet therapy in patients with acute non-cardioembolic ischaemic stroke associated with atherosclerosis, with no new safety signals, providing a rationale for testing these hypotheses in a phase 3 trial.

Despite observing a numerical reduction (albeit underpowered and non-significant) in recurrent symptomatic ischaemic stroke with highest dose of asundexian compared with placebo, there was no similar effect on incident covert MRI-detected brain infarcts. The absence of effect on covert infarction suggests that the mechanisms underlying these subclinical lesions are different from those causing most symptomatic ischaemic strokes, and further studies are required to help clarify this. Although previous work suggested that covert brain infarction could be a useful surrogate for assessing stroke therapies, its value in this regard has not been established.[24,25] Most incident covert brain infarcts were small subcortical lesions, presumed to result from underlying small-vessel disease, which might be unresponsive to anticoagulation treatment and impervious to FXI concentrations, as has been suggested in mendelian randomisation analyses.[12,13] FXIa inhibition in addition to antiplatelet therapy has not been previously assessed in patients with lacunar infarcts in the setting of small-vessel disease. Here, we sought to include a broad range of ischaemic stroke subtypes with the intention of exploring the effects of this new type of oral anticoagulant in patients with a variety of ischaemic

stroke mechanisms to inform the design of a subsequent phase 3 trial. One important result of this study is that covert brain infarcts (which were most often small and subcortical in our population) probably do not benefit from treatment with asundexian.

In this phase 2 trial, in which we included a broad spectrum of patients with non-cardioembolic ischaemic stroke, a heightened treatment effect for the composite post-hoc outcome of recurrent ischaemic stroke and transient ischaemic stroke was observed in patients with underlying atherosclerosis. Although these analyses were exploratory, several lines of biological plausibility support this finding, including (1) an a priori hypothesis based on the results of the COMPASS trial,[9] which found superior stroke prevention with combined FXa inhibition and aspirin therapy versus aspirin alone in patients with stable coronary and peripheral artery disease; (2) the consistently observed large effect size in PACIFIC-Stroke participants meeting TOAST criteria for large-artery atherosclerosis and those with any degree of atherosclerosis in extracranial or intracranial arteries; (3) a large and significant decrease in the occurrence of transient ischaemic attack with asundexian 20 mg and 50 mg daily versus placebo in post-hoc analyses, a cerebrovascular event that is substantially over-represented in patients with atherothrombotic disease;[26,27] and (4) the absence of reduction in the outcome of covert brain infarcts, the underlying pathology of which is largely due to cerebral small-vessel disease. Overall, the results of our post-hoc analyses are consistent with the hypothesis that anticoagulation with asundexian, when used in addition to antiplatelet treatment, reduces primarily stroke associated with atherosclerosis. FXI inhibition has been shown to reduce atherogenesis in experimental models of atherosclerosis,[28] and an interaction has been previously reported between dyslipidaemia and FXI concentrations, with a six-times increased risk of ischaemic stroke in patients with dyslipidaemia and increased circulating FXI concentrations compared with patients with dyslipidaemia without increases in FXI plasma concentrations and patients without dyslipidaemia.[29]

Four phase 2 randomised trials of therapies targeted at FXI or FXIa have found efficacy in preventing venous thromboembolism in patients undergoing total knee arthroplasty with associated low risk of bleeding in this postoperative setting.[17–20] Patients with severe congenital FXI deficiency rarely have spontaneous major bleeding, including intracranial haemorrhage. This is proposed to be because of the disproportionately large effect of FXI on thrombosis relative to its lesser ancillary role in haemostasis.[30] Accordingly, FXI and FXIa inhibitors have been postulated to effectively mitigate thrombosis with a relatively low risk of clinically relevant bleeding compared with conventional anticoagulants.[10] This hypothesis is supported by data from the PACIFIC-AF study,[10] with near complete inhibition of FXIa with asundexian, in which asundexian doses of 20 mg or 50 mg daily were associated with lower rates of bleeding than with apixaban in patients with atrial fibrillation.[10] In PACIFIC-Stroke, the absence of significant increases in the rates of major and clinically relevant non-major bleeding with asundexian treatment in patients with acute ischaemic stroke receiving antiplatelet therapy is also supportive of the safety of this new class of antithrombotic medication. Our findings are particularly reassuring because 43% of enrolled participants received dual antiplatelet therapy after randomisation, consistent with emerging evidence on improved secondary stroke prevention with short-duration dual antiplatelet therapy in patients with minor non-cardioembolic ischaemic stroke.[31] The excess numerical bleeding signal associated with asundexian on the primary safety outcome was driven by clinically relevant non-major bleeding. No increased risk of haemorrhagic transformation of the qualifying stroke was observed despite early post-stroke initiation of asundexian in this population.

The PACIFIC-Stroke phase 2 study was intended to inform the design of a subsequent phase 3 randomised trial, and as such has limitations related to statistical power for secondary outcomes and subgroup analyses. Notably, women comprised only 34% of participants. A substantial proportion of participants did not undergo serial study MRIs, requiring imputation of results for the outcome of incident covert brain infarction. Haemorrhagic transformation is more frequent with large infarcts, and despite eligibility allowing patients with NIHSS scores up to 15 to participate during part B, the cohort's mean NIHSS score at the time of randomisation was relatively low (mean 2·8), restricting the generalisability of our findings to patients with mild strokes. Furthermore, the post-hoc and exploratory nature of several of our notable findings should be interpreted with caution and require validation through future research.

In patients with acute non-cardioembolic ischaemic stroke of mild-to-moderate severity treated with antiplatelet therapy, asundexian did not reduce the composite primary outcome of symptomatic recurrent ischaemic stroke and incident covert brain infarcts on MRI in a dose-dependent manner. However, in post-hoc and exploratory analyses, we found that asundexian 50 mg daily provided superior protection against the composite of recurrent ischaemic stroke and transient ischaemic attack relative to placebo, especially in patients with atherosclerosis, without increasing the risk of major or clinically relevant non-major bleeding. These promising results require confirmation by an adequately powered phase 3 randomised trial.

Contributors

ASh and RGH wrote the first draft of the manuscript, and all coauthors subsequently reviewed the manuscript, provided comments, and approved submission. LX, LH, CN, and BK did the statistical analyses. EES directed the MRI Core Laboratory. ASh, HM, RGH, PC, and SJC accessed and verified the underlying data reported in the

manuscript. JM, IM, TH, AA, BC, VC, J-LM, QD, PT, HC, JMF, RV, RM, GMDM, TR, RL, ASt, SG, RR, LC, PK, JC, AGL, and AMD served on the steering committee, contributed to study design, recruited participants, and contributed to data collection and interpretation. The first and last authors (ASh and RGH) vouch for the fidelity of the trial protocol and for the accuracy and completeness of the data and reporting of adverse events and made the decision to submit the manuscript. All authors had access to data and accept responsibility for the decision to submit for publication.

Declaration of interests

All coauthors or their institutions received financial support from Bayer for participation in the PACIFIC-Stroke trial except HM, PC, BK, and CN who are employees of Bayer. HM, PC, BK, and CN do not hold any stock or stock options with Bayer.

Data sharing

The de-identified, individual participant-level data can be made available to investigators for secondary analyses after review of a submitted proposal by the PACIFIC-Stroke steering committee. Requests should be addressed to the corresponding author (ashkan.shoamanesh@phri.ca) or the sponsor representative (hardi.mundl@bayer.com). Requesters will be required to complete a study questionnaire. All requests will be assessed by the Steering Committee, who will review and comment on any potential publication of data from the trial.

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